

Conversational Healthcare Assistant

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1.0	30/11/2025	Kshitij , Ridham, Ipsita	Initial Document

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1. Problem Statement

Patients often face difficulty understanding medical reports, locating appropriate specialists, booking appointments, and receiving timely health reminders. The proposed solution provides an AI-driven, conversational experience that simplifies medical report interpretation, doctor discovery, and appointment management, thus enhancing patient engagement and outcomes.

2. Current State vs Target State

Current State:

- Fragmented portals
- Manual processes
- Limited automation

Target State:

- A unified chat interface for patients
- Automated medical report ingestion and analysis
- Personalized health insights powered by Azure OpenAI
- Integrated doctor search and appointment scheduling
- Email/SMS notifications

3. Context Diagram

Patient → Web/Mobile → Gradio → API Management → Chat Orchestrator → Azure OpenAI, Document Intelligence, Text Analytics, Provider APIs, Notification Services



4. Logical Architecture

Client Layer

- Web/Mobile application
- Conversational UI (chat widget)
- Document upload (PDF, image-based medical reports)

API & Orchestration Layer

- **Azure API Management (APIM)** for secure API exposure • **Chat Orchestrator**
 - **and Upload API:** ○ Handles conversation flow
 - Routes messages to Azure OpenAI
 - Manages document analysis and downstream workflows

AI Layer

- Azure OpenAI
- Document Intelligence
- Text Analytics

Data Layer

- Optional FHIR

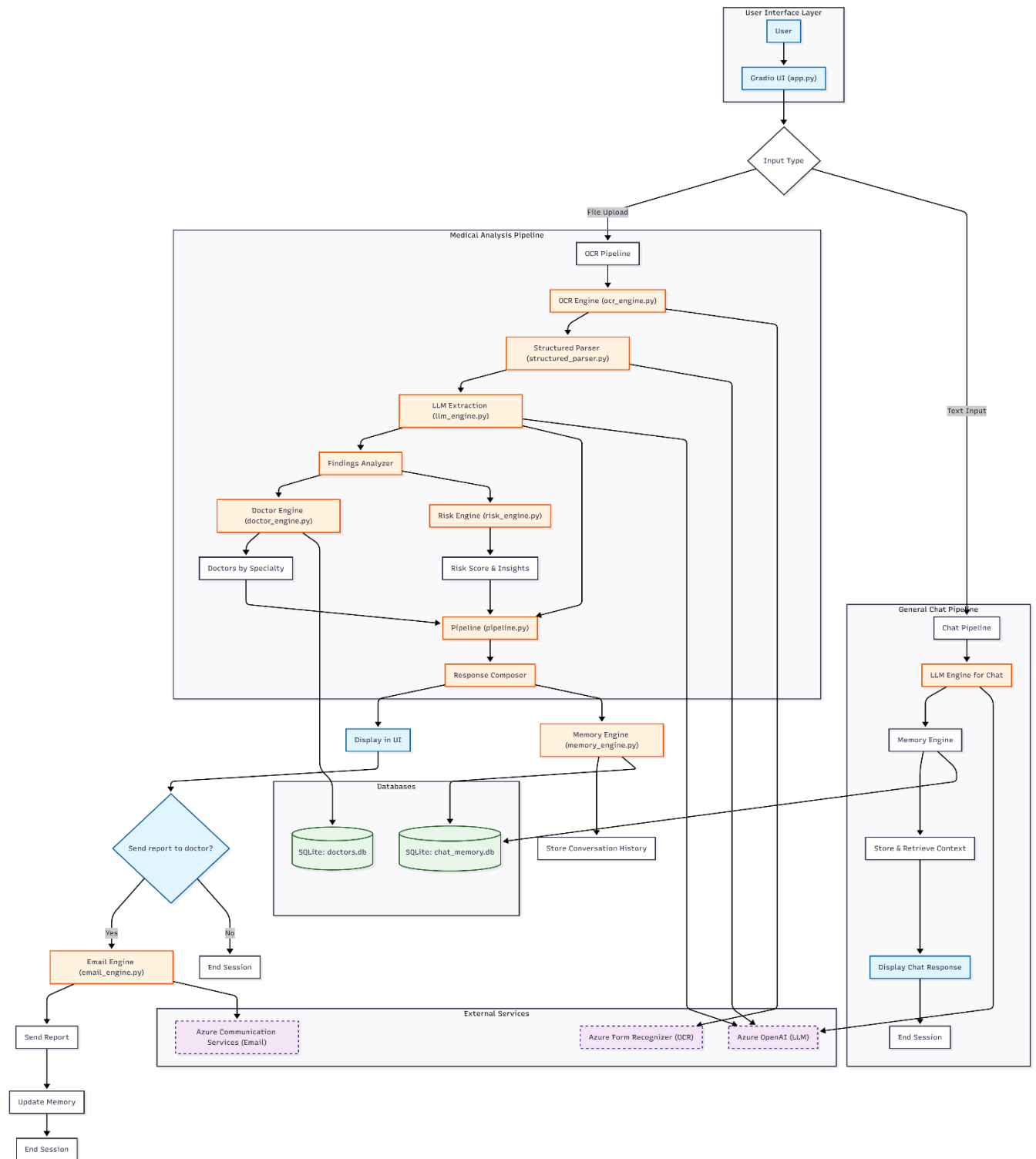
Integrations

- Provider APIs
- Notification systems

5. Deployment Architecture

Key Components:

- App Services/Container Apps
- APIM
- Azure OpenAI
- Cognitive Services
- Logic Apps
- Communication Services
- Key Vault



6. Data Flow

6.1 Conversational Flow

User sends message → APIM → Chat Orchestrator → Azure OpenAI

→ Generate response → Return to user

6.2 Report Upload & Analysis

User uploads report → Trigger Function

- Document Intelligence → Extract text/tables
- Text Analytics for Health → Identify conditions/medications
- Azure OpenAI → Generate summary & recommendations
- Store results and Notify users via chat

6.3 Doctor Search Flow

User requests doctor → Extract condition/location from chat

- Provider Data → Filter & rank results → Present options in chat

6.4 Appointment Booking Flow

- User decides to send an email to doctor
- doctor will reach out to patient as per their availability .
- Trigger notification workflow
- Return confirmation to user

6.5 Notification Flow

- Appointment created/updated
- Logic App triggered
- Azure Communication Services (Email)
- Send confirmation/reminder

7. Scalability & Reliability

- Autoscaling
- Retrying
- Circuit breakers
- Event-driven design
- Resilience patterns

8. Risks & Mitigations

Risk: AI hallucinations

Mitigation: -

- Prompt engineering

- grounding with retrieved data
- Content filtering

Healthcare Assistant
AI-Powered Medical Report Analyzer & Doctor Connect

Chat Session

how can you help me

I can help you understand your health questions, guide you through symptoms, explain medical terms, and give practical lifestyle or wellness advice. I can also help you make sense of lab reports or imaging results **once you upload them**, but I can't diagnose conditions or replace a healthcare professional.

Here are some ways I can support you:

- Explain what certain symptoms *might* mean
- Help you prepare questions for your doctor
- Offer diet, sleep, and stress-management tips
- Help interpret reports you share, in simple language
- Give general guidance on chronic conditions

If you tell me what you're dealing with or what you want to understand, I can tailor my help.

please analyse my health report
(Attached: GUR-0425-PA-0057162_Q-2025042717_250427_1635F.pdf_page_4.png)

Extracted Lab Findings

Test	Value	Unit	Range	Interpretation
25-hydroxy Vitamin D	21.7	ng/ml	20.0–30.0	low

what should I change in my diet because I have high risk score

Healthcare Assistant
AI-Powered Medical Report Analyzer & Doctor Connect

Chat Session

Test	Value	Unit	Range	Interpretation
25-hydroxy Vitamin D	21.7	ng/ml	20.0–30.0	low
Parathyroid Hormone	115.0	pg/ml	15.0–65.0	high

AI Clinical Evaluation

- Decision: consult
- Confidence: 0.83
- Rationale: Vitamin D deficiency with elevated PTH suggests secondary hyperparathyroidism, warranting specialist evaluation.
- Risk Score: 30/100**
 - 25-hydroxy Vitamin D: low
 - Parathyroid Hormone: high
- Suggested Doctors**
 - Dr. Kiran Yadav (Endocrinology) — 99999-5555

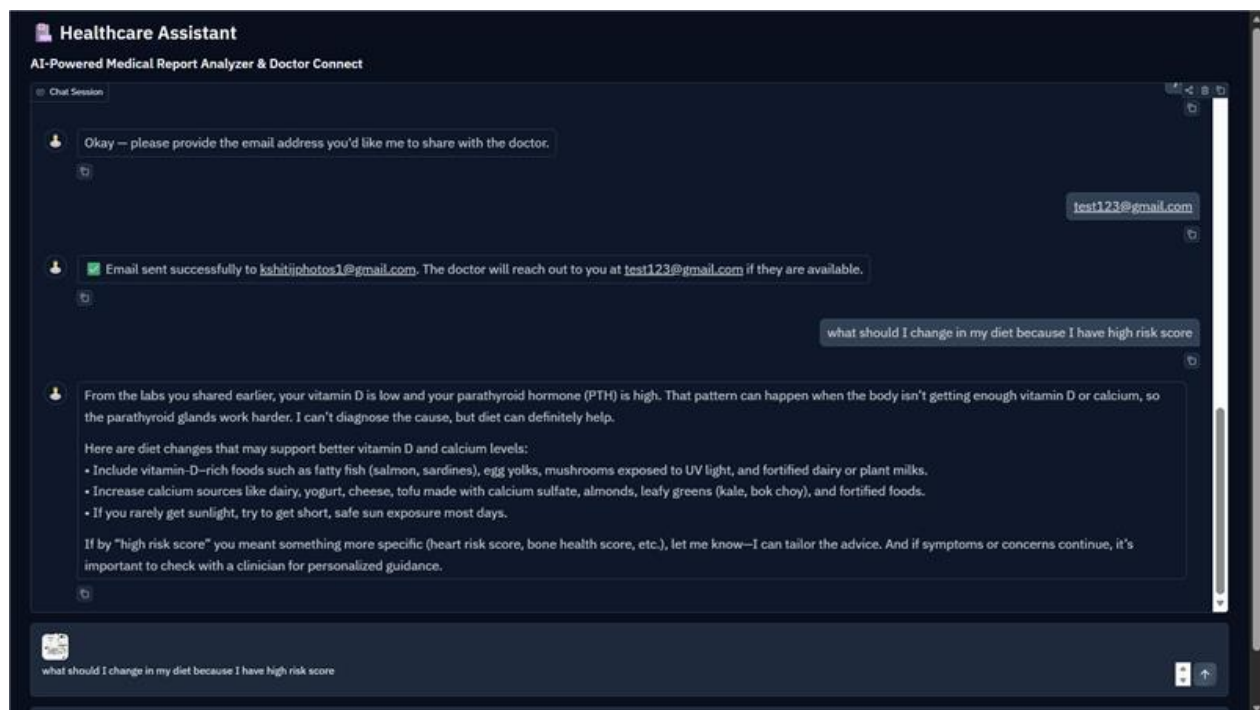
Would you like me to email this summary to the suggested doctor?

yes

Okay — please provide the email address you'd like me to share with the doctor.

test123@gmail.com

what should I change in my diet because I have high risk score





Kshitij Photos1 <kshitijphotos1@gmail.com>

Patient Report Review: Action Required

1 message

kshitijphotos1@gmail.com <kshitijphotos1@gmail.com>
To: kshitijphotos1@gmail.com

4 December 2025 at 15:37

Patient Consultation Request

Hello Doctor,

A patient has requested a review of their recent lab report. Please find the summary below.

Patient Contact: absc@gmail.com

Extracted Lab Findings

Test	Value	Unit	Flag
25-hydroxy vitamin D	21.7	ng/ml	low
parathyroid hormone	115.0	pg/ml	high

AI Clinical Analysis

Decision: CONSULT

Rationale: Markedly low vitamin D with elevated parathyroid hormone suggests possible secondary hyperparathyroidism, warranting specialist evaluation.

Note: This summary was generated by an AI assistant. Please review the full clinical context.



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Step to Access the Chatbot –

1. Extract the Zip Folder
2. Open Capstone healthcare project in a Code Editor .
3. Create a virtual env and install all the dependencies mentioned in requirements.txt .
4. Then run app.py and open the link mentioned in a terminal .
5. It will open a webpage , to interact with chatbot .
6. Upload your medical report and ask to analyze it .
7. If you want to schedule an appointment with doctor type yes in the prompt box when asked by the chatbot .